

Christopher O. Okoro

M.S. Data Science

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Innovative and results-driven Data Scientist with a solid engineering foundation and extensive experience building, training, and optimizing machine learning models to deliver measurable business value. Proven track record of developing end-to-end data pipelines, automation workflows, and interactive dashboards that transform complex, unstructured data into actionable insights. Adept at managing cloud deployments, designing ETL processes, and collaborating with cross-functional teams to solve real-world technical challenges and scale organizational footprint. **Work Authorization: U.S. Citizen (No sponsorship required).**

Qualifications Summary

- **Machine Learning & GenAI Expertise:** Skilled in building, training, and deploying predictive models, NLP frameworks, computer vision pipelines (YOLOv8/OCR), and Retrieval-Augmented Generation (RAG) solutions.
- **Data Engineering & ETL:** Proficient in designing scalable distributed data pipelines, cloud workflows (AWS/GCP), and web scraping scripts that significantly reduce manual data processing times.
- **Analytics & Visualization:** Expert at building KPI-driven dashboards (Tableau/Power BI) to translate complex data distributions into actionable strategic insights for executive stakeholders.
- **Cross-Functional Collaboration:** Strong history of leading onboarding processes, training team members, and aligning technical engineering specifications with strategic commercial initiatives.

Career Experience

Air Liquide, Katy, TX, United States
Data Analyst

June 2024 - November 2025

- Automated metadata extraction for 50,000+ engineering drawings by developing a two-stage classification pipeline using YOLOv8 object detection and PaddleOCR.
- Improved model accuracy by 15% mAP before division-wide deployment through dataset annotation, hyperparameter tuning, and computer vision optimization.
- Streamlined ETL processing for 30,000+ technical drawings by building Python and Pandas workflows integrated with AutoCAD scripts to extract and clean metadata.
- Increased operational efficiency for engineering teams by packaging and deploying the complete data validation pipeline as a standalone Windows application.
- Improved leadership visibility into project performance by building KPI dashboards that tracked throughput, processing velocity, and data quality metrics.

Exxon Mobil, Lagos, Nigeria
Data Analyst / Data Analyst Trainee

Sept 2022 – February 2024

- Automated the collection and preparation of live public datasets by building an end-to-end Python web scraping pipeline using BeautifulSoup to capture, clean, and transform data.
- Improved project visibility through real-time performance reporting by designing and deploying interactive Tableau dashboards with accurate project-tracking metrics.
- Reduced onboarding time by 20% for 10+ new team members by developing comprehensive technical frameworks and supporting data visualizations.

- Increased active user engagement by 40% by redesigning and managing the legacy corporate web application layer using ReactJS.
- Migrated client database consisting of 500,000 records by building a streamlined system architecture.
- Applied object-oriented design and modular UI practices across multiple active project deployments to optimize frontend framework scalability.

Projects

Distributed Data Processing Pipeline (Google Cloud Dataflow & Apache Beam)

- Built a scalable data pipeline using Apache Beam and Google Cloud Dataflow to ingest, transform, and process large datasets from GCS.
- Implemented comprehensive monitoring, logging, and resource tuning on GCP; produced complete infrastructure blueprints mapping GCS buckets, VM constraints, and Dataflow configurations.

Insurance Premium Forecasting & Classification (Machine Learning & Deployment)

- Engineered an end-to-end machine learning infrastructure (Logistic Regression, Random Forest, XGBoost) featuring rigorous EDA, balancing, and GridSearchCV parameter tuning.
- Deployed the top-performing statistical model as an interactive Streamlit web dashboard for live predictive assessment.

Education

Indiana University, Luddy School of Informatics, Computing, and Engineering, Bloomington, IN

- Master of Science in Data Science | Graduated May 2026 | 3.6/4.0
- Relevant Coursework: Statistics, Applied Machine Learning, Engineering Cloud Computing, Advanced Database Concepts, Big Data Management, Social Media Mining.

University of Lagos, Akoka, Lagos

- Bachelor of Science in Mechanical Engineering | Graduated October 2021

Leadership & Affiliations

- Mechanical Engineering Student Association, Public Relations Officer, UNILAG (Jan 2019 – Dec 2020).
- Nigerian Society of Engineers (NSE): General Member, Lagos, Nigeria (Jan 2020 – Present).

Certifications

- Google Advanced Data Analytics Specialization
- IBM Data Analyst Professional Certification

Technical and Analytical Skills

- **Data Science & Modeling:** Python (Pandas, NumPy, Scikit-learn, TensorFlow, PyTorch, Statsmodels), EDA, A/B Testing, Machine Learning, Computer Vision (YOLOv8, PaddleOCR), Retrieval-Augmented Generation (RAG).
- **Data & Cloud Engineering:** SQL, AWS/GCP (GCS, Dataflow), Apache Beam, BigQuery, Snowflake, Git, Docker, FastAPI, Database Migration.
- **Business Intelligence & Analytics:** Tableau, Power BI, Spreadsheets, Predictive Modeling, KPI Dashboards, Statistical Analysis.

Publications

- Christopher Okoro, et al. "Hybrid Renewable Power Plant Optimization with Electric Vehicle Charging System at the University of Lagos." AIAA 2021-3365. *AIAA Propulsion and Energy 2021 Forum*.